AAE 568 Applied Optimal Control and Estimation
3 credits

1 Course Information/Description

Optimal control is a time-domain method that computes the control input to a dynamical system which minimizes a cost function. The dual problem is optimal estimation which computes the estimated states of the system with stochastic disturbances by minimizing the errors between the true states and the estimated states. Combination of the two leads to optimal stochastic control.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the materials and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook: None. The course is based on a set of lecture notes and articles which will be made available during the class

3 Assignments

The course has a few homework assignments and a term project.

4 Course Learning Objectives

The main objective of this course is to study analysis and synthesis methods of optimal controllers and estimators for (stochastic) dynamical systems.

- Introduction of concept of optimal control and optimal state estimation and their applications
- Learn analysis and synthesis techniques for optimal estimation
- Analyze and design optimal controllers using calculus of variations and dynamic programing methods for both deterministic and stochastic dynamical systems

Main Topics

1. Review of some mathematical background
2. Classical optimal estimation
3. Optimal control
3. Stochastic optimal control and estimation

5. Evaluation

All assignments, project reports, etc., must be submitted on BS or Gradescope as instructed. No late homework will be accepted without the instructor's prior consent. Make sure to communicate with the instructor clearly and ASAP in case you anticipate a late submission.

Your grade will be calculated as follows:

- Homework Assignments: 30%
- Class Term Project: 70%